

Fundoo Notes UC8-UC16 Guide

This guide explains what was implemented from UC8 to UC16 with request examples.

Base URL

- `http://localhost:8080`

Common Header (Protected APIs)

- `Authorization: Bearer <JWT_TOKEN>`
 - `Content-Type: application/json`
-

UC8 - Note Creation (Safe User Mapping)

Endpoint

- `POST /api/notes`

Body

```
{
  "title": "My First Note",
  "content": "Hello world"
}
```

What happens

- Email is extracted from JWT `Principal` .
 - Server resolves current user from DB.
 - Note is saved with that user's `userId` .
 - Client cannot create notes for another user.
-

UC9 - Note Dashboard Fetch (IDOR-safe)

Endpoint

- `GET /api/notes`

What happens

- User identity is read from security context.
 - Only notes where `userId == currentUserId` and not deleted are returned.
 - No user id is accepted from request, preventing IDOR.
-

UC10 - Toggle APIs (Pin, Archive, Trash)

Endpoints

- `PUT /api/notes/{id}/pin`
- `PUT /api/notes/{id}/archive`

- `PUT /api/notes/{id}/trash`

What happens

- Server fetches note using `noteId + currentUserId` .
 - If note is not owned by user, operation fails.
 - Boolean state is toggled safely.
-

UC11 - AOP (Cross-cutting Concerns)

Implemented

- `LoggingAspect` with around advice.
- Logs method entry, execution time, and failures for controller/service layers.

Benefit

- Centralized logging and timing without duplicate code.
-

UC12 - RabbitMQ Messaging

Implemented

- Queue config.
- Producer publishes note-created event.
- Consumer listens and logs event.

Trigger point

- On successful note creation in `NoteService` .
-

UC13 - Redis Caching (JWT + OTP + TTL)

Implemented

- JWT cached at login with 1-hour TTL.
- JWT filter validates signature + token cache presence.
- OTP cache APIs with TTL 5 minutes.

OTP APIs

- `POST /api/cache/otp/send`

```
{ "otp": "123456" }
```

- `POST /api/cache/otp/verify`

```
{ "otp": "123456" }
```

UC14 - Spring Cache for Notes

Implemented

- `@Cacheable` on user notes fetch.
- `@CacheEvict` on create and toggle operations.

Result

- Faster repeated dashboard fetch for same user.
-

UC15 - JMS (ActiveMQ)

Implemented

- `JmsMessagingService` producer + listener.
- API to send JMS message:

Endpoint

- `POST /api/jms/send`

```
{ "message": "hello from jms" }
```

UC16 - Batch Excel Import

Implemented

- Excel parser service using Apache POI.
- Protected API to import notes from Excel file.
- Reads first sheet, expects:
 - Column A: title
 - Column B: content

Endpoint

- `POST /api/batch/notes/import` (multipart form-data)
 - Form key: `file`
-

Security fix for 403 on note POST

Done

- Stateless security session enabled.
- CORS support added.
- `OPTIONS` preflight permitted.
- JWT filter now accepts both:
 - `Authorization: Bearer <token>`
 - `Authorization: <token>`

Correct note create request

```
POST /api/notes
Authorization: Bearer <JWT_TOKEN>
Content-Type: application/json
```

```
{  
  "title": "My First Note",  
  "content": "Hello world"  
}
```

Quick Test Flow

1. Register: POST /api/users/register
2. Login: POST /api/users/login -> copy token
3. Create note: POST /api/notes
4. Fetch notes: GET /api/notes
5. Toggle pin/archive/trash
6. Test OTP cache
7. Test JMS send
8. Test Excel import